

SUMMARY TABLE: CHEMISTRY GRADE 10

Sub-Strand 3.1: Acids and Bases — Teacher Reference (Pre-filled)

SUMMARY TABLE: CHEMISTRY GRADE 10	
Sub-Strand	Sub-Strand 3.1: Acids and Bases
Driving Question	DRIVING QUESTION: "What invisible properties of everyday Kenyan liquids cause them to turn such different colours — and why does it matter for our health, our farms, and our environment?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Colour-Changing Drinks of the School Canteen — Launching the Phenomenon	Eight familiar Kenyan liquids — lemon juice, black chai, cow's milk, Stoney Tangawizi, tap water, baking soda solution, liquid handwash soap, and dilute Jik bleach — each turned a distinctly different colour when five drops of universal indicator solution were added: vivid reds and oranges for lemon juice, chai, and Stoney; green for tap water; and deep blues and purples for baking soda, soap, and Jik.	Substances can be classified as acids (sour taste, turn litmus red, turn universal indicator red/orange) or bases (bitter taste, slippery feel, turn litmus blue, turn universal indicator blue/purple). The colour differences produced by universal indicator are not random — they reveal a real, invisible property of each liquid. Many everyday Kenyan substances — lemon juice, Stoney Tangawizi, stomach acid (which helps digest ugali and githeri), toothpaste, soap, and antacids — are either acids or bases, and their uses depend on these properties. We never taste substances in a chemistry laboratory to test these properties.	The eight liquids turned different colours because they belong to two different categories of substances — acids and bases — which have different invisible properties that interact with the universal indicator dye to produce different colours. Liquids in the acid group (lemon juice, Stoney Tangawizi, black chai) turned red or orange because acids share a common property that triggers the red end of the indicator's colour range. Liquids in the base group (baking soda, liquid soap, Jik) turned blue or purple because bases share a different property that triggers the blue-purple end. Tap water, which is neither a strong acid nor a strong base, produced a stable green — the middle of the indicator's range. The model students built today can explain the groupings but not yet the differences in shade within each group, nor what exactly is dissolved in each liquid that causes it to behave as an acid or a base — those questions are what the rest of this unit will answer.
Lesson 2	Students watched an introductory video and	Acids are sour, turn litmus red, turn	Students cannot yet explain WHY acids and

What Are Acids and Bases? — Naming the Invisible	completed a paired reading on acids and bases. They observed that the eight canteen cups from the anchoring phenomenon (lemon juice, chai, Stoney Tangawizi, milk, tap water, baking soda solution, liquid soap, Jik bleach) can be systematically categorised as acid, base/alkali, or neutral based on their colours with universal indicator, litmus, phenolphthalein, and methyl orange.	phenolphthalein colourless, and turn methyl orange red. Bases (alkalis in solution) are bitter and slippery, turn litmus blue, turn phenolphthalein pink, and turn methyl orange yellow. Neutral substances (like clean tap water) do not change indicator colours in either direction. Common Kenyan substances — including Stoney Tangawizi, lemon juice, and stomach acid (which helps digest ugali and other foods) — are acids, while liquid soap, Jik bleach, and baking soda (used in mandazi) are bases. Indicators are tools that reveal whether a substance is acidic, basic, or neutral through predictable colour changes.	bases cause indicator colour changes at the particle level — this requires understanding what dissolved particles in acids and bases actually are, which will be investigated beginning in Lesson 3 through the Arrhenius definition and the concept of ions.
Lesson 3 What Is Inside an Acid or a Base? — The Arrhenius Definition	We observed — using diagrams and equations — that when HCl and H ₂ SO ₄ dissolve in water they release H ⁺ ions, and when NaOH and Ca(OH) ₂ dissolve in water they release OH ⁻ ions. We revisited the eight cups from Lesson 1 and connected their colours to the presence of these specific ions.	We learned the Arrhenius definitions: an acid produces H ⁺ ions in aqueous solution; a base produces OH ⁻ ions in aqueous solution. We learned to write ionisation equations for HCl, H ₂ SO ₄ , NaOH, and Ca(OH) ₂ , and we learned the terms monoprotic and diprotic.	We can now explain why the cups in Lesson 1 turned different colours: the cups that turned red or orange contained substances that release H ⁺ ions (like lemon juice and Stoney Tangawizi), while the cups that turned blue or purple contained substances that release OH ⁻ ions (like liquid soap and Jik). The Arrhenius model gives us particle-level explanatory power over the phenomenon we observed on Day 1.
Lesson 4 Differences Between Acids and Bases Using Commercial Indicators	Students tested seven familiar Kenyan liquids — lemon juice, vinegar (siki), baking soda solution, liquid soap solution, tap water, black tea (chai), and cow's milk — against red litmus, blue litmus, phenolphthalein, and methyl orange indicators. They recorded specific colour changes for each solution-indicator combination and classified each liquid as acid, base, or neutral.	Acids turn blue litmus red (red litmus stays red), turn phenolphthalein colourless, and turn methyl orange red. Bases turn red litmus blue (blue litmus stays blue), turn phenolphthalein pink/magenta, and turn methyl orange yellow. Neutral substances produce no litmus colour change, leave phenolphthalein colourless, and leave methyl orange orange-yellow. Common Kenyan everyday liquids can be reliably classified using these indicator colour rules.	The colour changes produced by commercial indicators in acidic and basic solutions follow a consistent, reproducible pattern. This pattern allows scientists — and students — to classify any unknown solution as an acid, a base, or neutral without tasting or smelling it. The familiar cups from the school canteen phenomenon can now be classified by their indicator behaviour, revealing that invisible dissolved particles in everyday liquids have measurable, testable properties. However, today's experiment tells us only WHAT a substance is (acid or base), not HOW STRONGLY acidic or basic it is, nor HOW it will react chemically — these questions drive the next phase of our investigation.
Lesson 5 Chemical Properties of Acids: Reactions with	Students observed: (1) vigorous or moderate bubbling when dilute HCl and H ₂ SO ₄ were added to zinc and magnesium	Students learned that: (1) acids react with metals (zinc, magnesium) to produce a salt and hydrogen gas — the balanced	The colour-changing canteen cups from the anchoring phenomenon now reveal more than just indicator colours — the cups

Metals and Neutralisation	ribbon; (2) a 'pop' sound confirming the production of hydrogen gas when a lit splint was held at the mouth of the test tube; (3) the solution in the acid + metal test tube becoming slightly warm; (4) phenolphthalein indicator remaining colourless in dilute HCl, then turning pink permanently at the endpoint when dilute NaOH was added dropwise; (5) the corroded whitish-grey deposits on a car battery terminal and rust on a corrugated iron roofing sheet — both connected to acid reactivity.	molecular equation for $\text{Zn} + \text{HCl}$ is $\text{Zn(s)} + 2\text{HCl(aq)} \rightarrow \text{ZnCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ and the net ionic equation is $\text{Zn(s)} + 2\text{H}^+\text{(aq)} \rightarrow \text{Zn}^{2+}\text{(aq)} + \text{H}_2\text{(g)}$; (2) hydrogen gas is identified by the 'pop' sound of a lit splint — not by sight alone; (3) acids react with bases in a neutralisation reaction to produce a salt and water — the net ionic equation $\text{H}^+\text{(aq)} + \text{OH}^-\text{(aq)} \rightarrow \text{H}_2\text{O(l)}$ shows it is the H^+ and OH^- ions that react; (4) phenolphthalein is a reliable endpoint indicator for acid–base neutralisation; (5) the chemical reactivity of acids with metals explains real-world phenomena such as corroded battery terminals and structural metal degradation.	containing acidic liquids (lemon juice, Stoney Tangawizi, black tea) contain H^+ ions that can react with metals and be neutralised by OH^- ions. The H^+ ion is the active particle responsible for both the colour changes seen in Lessons 1–4 AND the chemical reactivities observed today. The corroded car battery terminal is explained by H_2SO_4 (battery acid) reacting with the metal terminals via the acid + metal reaction. Neutralisation explains why antacids (bases) relieve stomach acidity — the OH^- or carbonate ions in the antacid neutralise the excess HCl produced by the stomach, reducing H^+ concentration and relieving discomfort. Every 'invisible property' we uncover brings us closer to a complete answer to the driving question.
Lesson 6 Acid Reactions with Carbonates and Hydrogen Carbonates — Completing the Acid Reaction Picture	Students observed that: (1) when dilute HCl was added to marble chips (CaCO_3), vigorous fizzing/effervescence occurred at the marble surface, the marble chips gradually dissolved and became smaller, and the gas produced turned limewater from clear to milky white; (2) when dilute HCl was added to baking soda (NaHCO_3), very rapid effervescence occurred (faster than with marble chips), the powder dissolved quickly into solution, and the gas again turned limewater milky white; (3) in both cases the limewater test confirmed the gas was carbon dioxide (CO_2).	Students learned that: (1) acids react with metal carbonates to produce a salt, water, and carbon dioxide gas — e.g., $2\text{HCl} + \text{CaCO}_3 \rightarrow \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2$; (2) acids react with metal hydrogen carbonates to produce a salt, water, and carbon dioxide gas — e.g., $\text{HCl} + \text{NaHCO}_3 \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{CO}_2$; (3) the standard test for CO_2 is to bubble the gas through limewater — a milky precipitate of calcium carbonate (CaCO_3) confirms CO_2; (4) these reactions explain why Kericho farmers add lime to acidic soils (neutralises excess acid, releases CO_2) and why battery acid (H_2SO_4) can corrode carbonate-containing materials at Mombasa port; (5) combining evidence from Lessons 4, 5, and 6, acids react with four types of substances — metals, metal oxides, metal carbonates, and metal hydrogen carbonates — each producing a salt, and each producing specific additional products (H_2 gas, water, or water + CO_2).	Students can now explain that: (1) H^+ ions from the acid attack the carbonate ion (CO_3^{2-}), forming unstable carbonic acid (H_2CO_3) which immediately decomposes into water and carbon dioxide gas — this is why fizzing is observed; (2) the more reactive or finely divided the carbonate, the faster the reaction (explaining why powdered NaHCO_3 reacts faster than solid marble chips); (3) the CO_2 gas produced during the Stoney Tangawizi carbonation process is related to the same chemistry — carbonic acid (H_2CO_3) dissolved under pressure; (4) returning to the canteen phenomenon: the acidic cups (lemon juice, Stoney Tangawizi, black tea, and the soft drink) would all react with carbonates and hydrogen carbonates — explaining both their indicator colours and their real-world chemical behaviour in contexts like digestion, agriculture, and industry.
Lesson 7 Consolidation of Chemical Properties of Acids and Bases — Building the Reaction Summary Model	In the Gallery Walk (Phase 2), students revisited evidence from all six prior lessons: the rainbow of indicator colours produced by the eight canteen liquids (lemon juice = red, Stoney Tangawizi = orange-red, chai = orange, milk = yellow-green, tap water =	Students learned that all four types of acid reactions — acid + metal, acid + carbonate, acid + metal oxide, and acid + alkali (neutralisation) — can be unified under the Arrhenius definition: in every case, H^+ ions from the acid are the active chemical agent.	Students constructed a Graphic Summary Model synthesising all chemical properties of acids and bases onto one structured visual. They explicitly explained that the eight canteen cups turn different colours because they contain different

	green, baking soda = blue, liquid soap = blue-violet, Jik = purple); the results of reactions of HCl with zinc metal (effervescence, squeaky pop confirms H_2), with marble chips/sodium carbonate (effervescence, limewater turns milky confirms CO_2), with copper oxide (black solid dissolves in blue solution), and with NaOH solution (no visible change — neutralisation); and the indicator behaviour of bases. In Phase 3, the teacher demonstrated the CO_2 limewater test and the H_2 burning splint test live, adding fresh direct observation to consolidate prior learning.	They learned the generalised word and symbol equations and observable evidence for each reaction type, including the specific gas tests. They learned that bases are characterised by OH^- ions (Arrhenius), react with acids in neutralisation ($H^+ + OH^- \rightarrow H_2O$), and produce distinctive colour changes in indicators opposite to those of acids. Students consolidated knowledge that the colour of each canteen cup in the phenomenon is causally determined by the concentration of H^+ or OH^- ions — not by appearance, taste, or smell alone.	concentrations of H^+ or OH^- ions: liquids with high H^+ concentration (lemon juice, Stoney Tangawizi) produce red/orange colours because they have low pH; liquids with high OH^- concentration (Jik, liquid soap) produce blue/purple colours because they have high pH; tap water, with equal H^+ and OH^-, produces green (neutral, pH 7). Students explained the neutralisation reaction between stomach acid (HCl, producing H^+) and antacid medication (a base, producing OH^-) as a real-world application. They updated the Driving Question Board to mark Column 2 (chemical properties) as substantially answered, and generated new questions about acid/base STRENGTH targeting Lessons 8–9.
Lesson 8 Introduction to pH: Measuring the Strength of Acids and Bases in Kenyan Everyday Liquids	Students measured the pH of eight Kenyan everyday liquids using pH indicator paper and universal indicator solution, and recorded the colour changes and approximate pH values for each solution. They also received reference pH data for six additional Kenyan real-world liquids including stomach acid, lime juice used in Mombasa cooking, Lake Magadi water, and Kericho tea farm soil runoff.	Students learned that pH is a numerical scale from 0 to 14 that measures the relative concentration of H^+ ions in a solution. A pH below 7 indicates an acidic solution, pH 7 indicates a neutral solution, and pH above 7 indicates a basic solution. Each step on the pH scale represents approximately a tenfold change in H^+ ion concentration. Universal indicator shows a continuous colour spectrum that allows finer discrimination between solutions than litmus alone.	Students used their pH data to explain why the eight canteen cups from the Lesson 1 phenomenon turned different colours: each colour corresponds to a specific pH value that reflects the relative concentration of H^+ ions in that liquid. They connected pH measurement back to the Arrhenius definition by explaining that substances with more H^+ ions in solution have lower pH values and produce redder colours with universal indicator, while substances with more OH^- ions have higher pH values and produce bluer or violet colours. They updated the Driving Question Board with a new focused question about what the pH number reveals about the strength — not just the presence — of an acid or base.
Lesson 9 Strong vs. Weak Acids and Bases: Why the Same pH Step Can Make or Break a Tea Farm	From our Lesson 8 pH data, two acids at the same concentration gave different pH readings: hydrochloric acid gave pH 1 while ethanoic acid (vinegar) gave approximately pH 3, and two bases at the same concentration gave different pH readings: sodium hydroxide gave pH 13 while ammonia solution gave approximately pH 11. These differences could not be explained by concentration alone.	Strong acids (such as HCl and H_2SO_4) and strong bases (such as NaOH and $Ca(OH)_2$) ionise completely in aqueous solution, producing a high concentration of H^+ or OH^- ions and therefore giving pH values far from 7. Weak acids (such as ethanoic acid in vinegar, citric acid in lemon juice, and carbonic acid in Stoney Tangawizi) and weak bases (such as ammonia solution and soap) ionise only partially, producing fewer H^+ or OH^- ions and giving pH values closer	The different colours produced in the eight cups of the anchoring phenomenon reflect not only whether a substance is an acid or base, but also how strongly it ionises. The lemon juice turned orange-red rather than deep red because citric acid is a weak acid that ionises partially, producing fewer H^+ ions and a higher pH than a strong acid at similar concentration. The Jik bleach turned deep purple-blue because NaOH is a strong base that ionises completely, producing a

		to 7. The degree of ionisation is shown by a one-directional arrow for strong acids and bases and a reversible double arrow for weak acids and bases.	very high OH ⁻ concentration and a pH well above 7. In Kericho, soil acidity caused by weak organic acids is still damaging at pH 4.2 because even partial ionisation of many acid molecules accumulates enough H ⁺ ions to harm tea plant roots, and farmers apply calcium hydroxide (a strong base) to neutralise this acidity through the neutralisation reaction studied in Lesson 5.
Lesson 10 Uses of Acids and Bases + Final Model Build & Explanation	Students watched a video showing Kenyan contexts — a Kericho tea farmer applying agricultural lime to acidic soil, a Nairobi garage mechanic using sulphuric acid in a car battery, a Mombasa pharmacist describing antacid tablets, a soft-drinks factory producing carbonic acid in Stoney Tangawizi, and a Kisumu soap-making cooperative using sodium hydroxide. Students also observed the eight-cup universal indicator demonstration (via projected image callback from Lesson 1), noting the full colour spectrum from red (lemon juice, pH ~2) through green (tap water, pH ~7) to purple (Jik bleach, pH ~12–13).	Students learned that acids and bases have wide practical uses across Kenyan life: hydrochloric acid (pH ~1–2) aids digestion in the stomach; carbonic acid gives carbonated drinks like Stoney Tangawizi their fizz (pH ~3–4); sulphuric acid powers car batteries; calcium hydroxide (agricultural lime) neutralises acidic soils in the Rift Valley to improve crop yields; sodium hydroxide is used in soap manufacture; and bases such as sodium bicarbonate and magnesium hydroxide are active ingredients in antacid medications that neutralise excess stomach acid. Students consolidated the Arrhenius definitions (acids release H ⁺ ; bases release OH ⁻), the pH scale (0–14), and the relationship between pH and universal indicator colour, and applied these to explain every colour in the eight cups from Lesson 1.	Students explained that the colour differences among the eight cups arise because each liquid contains dissolved substances that release different amounts of H ⁺ or OH ⁻ ions in water. Liquids with high H ⁺ ion concentration (low pH) — such as lemon juice, Stoney Tangawizi, and chai — cause universal indicator to turn red, orange, or yellow. Tap water, which is neutral (pH ~7), produces a stable green colour. Liquids with high OH ⁻ ion concentration (high pH) — such as baking soda solution, liquid soap, and Jik bleach — cause universal indicator to turn blue or purple. This matters for health (stomach acid, antacids), for farming (soil pH and lime application), and for the environment (industrial effluent pH and water quality in rivers like the Nairobi River).

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